MILITARY SPECIFICATION

ANTENNA AS-515A/SPG and MX-7232/SPG-53B

(RADAR ANTENNA SCANNER)

This specification is approved for use by the Naval Sea Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification establishes requirements for the procurement of radar antenna scanning units designated as Antenna AS-515A/SPG (Vertically polarized) and MX-7232/SPG-53B (Horizontally polarized). They are associated with various antenna mounts and reflectors which are not part of the radar antenna scanner. See 6.2.1.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-S-901	Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper
MIL-P-15024	Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-L-17672	Lubricating Oil, Hydraulic and Light Turbine, non-corrosive
MIL-F-18870	Fire Control Equipment, Naval Ship and Shore, General Specification

STANDARDS

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Naval Sea Systems Command (Code Ident. 10001)

LD-477136	Antenna AS-515A/SPG and MX-7232/SPG-53B Radar Scanners (including all drawings and specifications listed thereon)

PUBLICATIONS

Naval Sea Systems Command

NAVORD OP 1600	Gun Fire Control System MK 56 with Radar Equipment MK 35, Description, Installation, Operation and Maintenance

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NAVORD OP 2132	Radar Equipment MK 34
NAVORD OP 3682	Radar Set AN/SPG-34, Description, Installation, Operation and Maintenance
NAVORD OP 2166	Radar Set AN/SPG-50, Description, Installation, Operation and Maintenance
NAVORD OP 2782	Radar Set AN/SPG-53A, Description, Installation, Operation and Maintenance
NAVORD OP 2782 SUPP 1	Radar Set AN/SPG-53B for GFCS MK 68 Mods 9 and 10, Description, Operation and Maintenance
NAVORD OP 2782 SUPP 4	Radar Set AN/SPG-53C for GFCS MK 68 Mod 12, Description, Operation and Maintenance
NAVORD OP 2782 SUPP 7	Radar Set AN/SPG-53D for GFCS MK 68 Mod 14, Description, Operation and Maintenance

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The antenna scanner, when operating as an integral part of a radar set, shall radiate R.F. pulses, generated by the radar transmitter, as a narrow beam. The complete antenna assembly shall be comprised of a feedhorn and a drive motor and mechanism, capable of causing the feedhorn to nutate in such a manner so as to describe either a spiral or circular path at its radiating end. This motion, in turn, shall cause the radiated beam to scan in a spiral or circular path. The mechanism shall include a solenoid-operated clutching device to select the mode of scan (spiral or conical), and two reference voltage generators to indicate the angular position and the amount of offset of the feedhorn during its scanning cycle. The components (except the feedhorn) shall be encased in a watertight housing which shall be capable of being flange-mounted at the center of a paraboloidal reflector in such a manner that the radiating end of the feedhorn will describe its spiral or circular path about the focal point of the reflector, in a plane perpendicular to the focal axis of the reflector.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the antenna scanner shall be manufactured using the methods and procedures proposed for production. See 6.2.1. The samples shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. Further production of antenna scanners by the supplier prior to the approval of the preproduction sample shall be at the supplier's risk.

3.3 Components. Each antenna scanner shall consist of the components specified by the drawings listed on LD-477136, assembled to provide a complete functional unit.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified by documents invoked herein shall be selected in accordance with MIL-STD-143.

3.5 Materials.

3.5.1 Lubricants. Lubricating oil shall be Military Symbol 2135 T-H per MIL-L-17672 Lubricating Oil, Hydraulic and Light Turbine, Noncorrosive. This is equivalent to SAE-20 commercial grade.

3.5.2 Standard parts. Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be as identified on the drawings. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by standard parts (MS, AN or JAN) without alteration, and

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provided the corresponding standard part numbers are referenced in the parts list or on the drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to applicable requirements of this specification. See 6.2.2.

* 3.6 Design and construction. The antenna scanner shall be manufactured in accordance with MIL-F-18870, and the applicable drawings listed on LD-477136. Minor variations of component configuration are permitted for undimensioned design detail, provided applicable requirements of this specification are met. In the event that a discrepancy exists between the drawings and this specification the latter shall take precedence. See 6.2.2.

3.6.1 Electrical circuits. All electrical circuits (except reference voltage generator circuits) shall be capable of withstanding a potential of 200 percent of operating voltage plus 1000 volts root-mean-square (RMS) for a period of 5 seconds without breakdown or damage to the insulation.

3.6.2 Wiring. Hook-up wire employed shall have adequate protection against abrasion. Skinned ends of braid covered wires shall have the insulation covered with air drying varnish or lacquer for a distance of one-half inch back from the bare wires, to prevent fraying.

3.6.3 Terminal lugs. Terminal lugs shall be either soldered, or the solderless type in accordance with MIL-T-7928. Where solder-type lugs are used, connections shall be clean and mechanically secure before soldering. The ferrule portion of the terminal lugs shall be covered by snugly-fitting flexible plastic tubing which shall extend at least one-fourth inch over the insulation of the wire.

3.7 Performance.

3.7.1 Electrical.

3.7.1.1 Drive Motor. The drive motor shall be a 3-phase, 60-hertz (Hz), 440-volt (V), induction-type motor with an adequate horsepower rating to meet the mechanical load requirements with balanced phase conditions. The nominal speed shall be 1735 revolutions per minute (RPM) with a minimum full load speed of 1700 RPM. The rated load current drawn by any phase of the motor shall be no greater than 1.25 amperes (A) when operating from a balanced 3-phase line voltage of 440 ± 5 V. Any difference (imbalance) between phase currents shall be within 5 percent of the largest value of current drawn by one phase. The motor shall be capable of starting under the required load, and accelerating to its minimum full load speed in less than 5 minutes under any of the ambient conditions specified in 3.8.1. Continuous operation from a 440 V ± 10 percent, 60 ± 3 Hz source, shall be possible without malfunction or damage to the equipment. A voltage surge of 440 V + 30 percent for one second shall not damage the equipment and it shall not require more than 10 seconds for recovery to normal operating conditions. Temperature rise under any condition of operation shall not exceed 160°F (71.1°C).

3.7.1.2 Heater and thermostat. To insure starting and operating at low temperatures, the antenna scanner shall be equipped with a heating device which shall maintain the temperature within the motor and mechanism encasement at an operable level. The heater shall be controlled by a thermostatic switch set to operate at a temperature of 45 ± 5°F, to limit the heater operation to housing internal temperatures below this setting. The heater shall be rated at 150 ± 15 watts (W), and is to be energized from a 120 V ± 10 percent 60 ± 3 Hz source.

* 3.7.1.3 Reference voltage generators. Two 2-phase reference voltage generators, which are part of the mechanism, shall produce, during conical scan, a sinusoidal 29 ± 2 Hz 115 ± 10 volts-per-phase output at nominal motor speed. The phase output voltage shall contain less than 5 percent RMS total harmonic content when connected to a 50,000-ohm resistive load. The two phases of each generator shall be 90 ± 2 electrical degrees apart. The internal resistance of each phase (with compensating resistors added if necessary) shall be 1500 ohms ± 10 percent. Alinement and phasing of the generators with respect to the position of the feedhorn shall be as shown on Drawing 1357736.

3.7.1.4 Solenoid. The clutch solenoid shall normally operate from a 115 V, 60 Hz source, but shall be capable of positive operation at any voltage from 103 to 127 volts at 60 ± 3 Hz.

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* 3.7.1.5 RF components. The waveguide components for the antenna scanner, including the feedhorn, shall operate over the frequency band of f_L to f_L + 1100 MHz at a peak power of 500 kW. "f_L" is defined as the low frequency limit of the operating band delineated in OP 1600, 2132, 3682, 2166, or 2782, as applicable. The voltage standing wave ration (VSWR) at the antenna input for the assembled radio frequency (RF) components, including the feedhorn but without the antenna reflector mounted, shall not exceed the following, for the frequencies indicated:

TEST FREQUENCY (MHz)	VSWR	% REFLECTION (VOLTAGE)
f_L + 0	1.45	18.50
f_L + 250 ± 20	1.3	13.0
f_L + 550 ± 20	1.3	13.0
f_L + 850 ± 20	1.3	13.0
f_L + 1100	1.3	13.0

3.7.2. Mechanical

* 3.7.2.1 Conical scan. In conical scan, the feedhorn shall nutate at such an angle that the circle described by a point in the center of the feedhorn cap shall be 0.5 + 0.015, - 0.045 inch in diameter. The concentricity of this circle with respect to the mounting pilot shall be within 0.05 inch.

* 3.7.2.2 Spiral scan. In spiral scan the feedhorn shall nutate at such angles that a point in the center of the feedhorn cap shall describe a spiral between the circle of conical scan and a circle 4.00 ± 40 inches in diameter. The rate of spiral shall be approximately two hertz.

3.7.2.3 Scan selection time. The time required to shift from spiral to conical scan (or vice versa) shall not exceed the time required for one complete cycle of spiral scanning.

3.7.2.4 Nutation speed and direction. Nutation for both types of scan shall be at motor speed and in a counterclockwise direction as viewed from the radiating end of the antenna.

* 3.7.3 Lubrication. All moving parts shall be lubricated by oil dispersed throughout the housing by means of an oil pump. The oil pump shall be such that priming shall not be required. A lubrication instruction plate per Dwg 1031518 shall be affixed to the rear housing cover in the vicinity of the oil filler pipe. Refer to para 3.5.1 for the proper lubricant.

3.8 Environmental.

3.8.1 Temperature. The antenna scanner shall be capable of starting and operating normally in the temperature range of -20°F to +160°F. The antenna shall be suitable for storage, without damage, in any ambient temperature from -40°F to +160°F.

3.8.2 Degrees of enclosure. The antenna scanner shall be watertight and meet the requirements of 3.1 after it has been tested in accordance with 4.10.3.

3.8.3 Shock. The antenna scanner shall show no evidence of damage and shall meet the requirements of 3.1 after it has been tested in accordance with 4.10.4.

* 3.9 Identification plate. An identification plate in accordance with MIL-P-15024 shall be affixed to the upper portion of the rear housing cover plate. Serial numbers shall be as assigned by NAVSEASYSCOM. See 6.2.1.

3.10 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a workmanlike manner to assure compliance with all requirements of this specification and drawings listed in LD-477136. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, machine screw assemblage, freedom of parts from burrs and sharp edges, and freedom of gaskets and fittings from oil leaks.

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* 3.11 Interchangeability. The design of the antenna scanner shall be such that all parts are interchangeable between antenna scanners, with the exception of the following five parts:

Rotor Shaft (front)	Dwg. 458962
Rotor Shaft (center)	Dwg. 597641
Rotor Shaft (rear)	Dwg. 597644
Control Shaft	Dwg. 458960
Waveguide Support Tube	Dwg. 1357748

These parts are to be assembled and dynamically balanced at final assembly. After balancing, each of the five parts shall be marked with a correspondeng serial number. Although individual parts are not interchangeable, the assembly consisting of the five parts shall be interchangeable from scanner to scanner.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requriements.

* 4.2 Classification of inspection. Inspections of the antenna scanner assembly shall be classified as follows:

(a) Preproduction inspection. (See 4.3)

(b) Periodic production inspection. (See 4.4)

(c) Quality conformance inspection. (See 4.5)

* 4.3 Preproduction inspection. The preproduction sample, after satisfactorily passing the quality conformance inspection of 4.5, shall be subjected to the following tests and examinations, in the sequence listed, at the activity designated by the procuring activity. See 6.2.1, 6.2.2. Acceptance of the preproduction sample shall be based on no defects in the sample for the following tests:

Paragraph	Examination
4.7	Visual
4.10.1	Temperature
4.10.1.1	Storage
4.10.1.2	Low temperature starting
4.10.1.3	Operating
4.10.2	Vibration
4.10.3	Degree of enclosure
4.10.4	Shock

4.3.1 Preproduction samples. Preproduction sample size shall be as specified in the contract or purchase order. See 6.2.1.

* 4.4 Periodic production inspection. The periodic production sample selected in accordance with 4.4.1 shall be subjected to the tests and examinations of 4.10 at the activity designated by the procuring activity. Failure of periodic production samples to pass the tests of 4.10 shall be cause for withholding approval of all antenna scanners in the lots from which the samples were drawn. See 6.2.2.

4.4.1 Periodic production samples. Periodic production samples shall be selected from antenna scanners that have satisfactorily passed the quality conformance inspection of 4.5. Periodic production sample size shall be as specified in the contract or purchase order. See 6.2.1.

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4.5 Quality conformance inspection. Each antenna scanner shall be subjected to the following tests and examinations. Failure of the antenna scanner to pass any test specified shall be cause for rejection. See 6.2.2.

Paragraph	Examination
4.7	Visual
4.8	Electrical tests
4.8.1	Variation in supply line voltage
4.8.2	Temperature rise in motor
4.8.3	Dielectric strength
4.8.4	Dielectric strength of reference voltage generators
4.8.5	RF components
4.9	Mechanical tests

4.6 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.7 Visual inspection. The antenna scanner shall be examined for conformance to the requirements specified in this specification and drawings listed on LD-477136. The antenna scanner shall also be examined for workmanship in accordance with 3.10.

4.8 Electrical tests. The antenna scanner shall be connected to power sources of nominal voltages and frequencies for all electrical components as specified in 3.7.1. With the equipment in operation, electrical measurements shall be made with instruments of the proper type, range, and calibration to determine whether the following requirements have been met:

(a) The input current to each phase of the antenna scanner motor specified in 3.7.1.1.

(b) With the thermostat switch bypassed, the power input to the heater specified in 3.7.1.2.

(c) The output of each reference voltage generator specified in 3.7.1.3.

(d) The internal resistance of each reference voltage generator specified in 3.7.1.3.

(e) The phase of the reference voltage generators shall be measured with reference to each other and to the center of the emitting end of the feedhorn while the antenna scanner is operating in conical scan. The phase adjustments shall be in accordance with Dwg. 1357736.

(f) Clutch solenoid operation specified in 3.7.1.4.

4.8.1 Variation in supply line voltage. Electrical components of the antenna scanner shall be operated with supply line voltages varied between the limits specified in 3.7.1 to determine compliance with the requirements specified therein.

4.8.2 Temperature rise of motor. The temperature rise of the antenna scanner drive motor shall be determined by the following test:

Place antenna scanner in an area or temperature-controlled room where ambient temperature can be maintained at ± 2°F throughout the test. The mean ambient temperature shall be above 60°F and preferably at 75°F. During the test the antenna scanner shall be protected from drafts and heat radiation from outside sources. Temperature rise of the antenna scanner motor shall be determined by the resistance method and the following formula shall apply:

$$\text{Temperature rise, °C} = \frac{(234.5 + tc)\,Rh}{Rc} - (234.5 + ta)$$

Where:

Rc = Cold resistance of winding

Rh = Hot resistance of winding

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tc = Temperature (°C) of motor when cold resistance was measured. This ambient temperature shall have been maintained for not less than 24 hours immediately preceding the cold resistance measurement.

ta = Ambient temperature (°C) during the last quarter of the test.

Constant temperatures shall be considered to have been reached when four consecutive readings taken at 15-minute intervals indicate no increase in temperature. The allowed temperature rise of the motor is specified in 3.7.1.1.

4.8.3 Dielectric strength. All electrical circuits in the antenna scanner, except those pertaining to the two reference voltage generators, shall be subjected to a high potential test of 200 percent operating voltage plus 1000 V rms. The voltage shall be increased slowly from 0 to the value specified above (in about 5 seconds), held at this value for not less than five seconds, and then gradually decreased to 0. Examinations for evidence of insulation breakdown shall be conducted during this test.

4.8.4 Dielectric strength of reference voltage generators. The reference voltage generators shall be subjected to a high potential test of 740 ± 20 V. The voltage shall be increased slowly from 0 to 740 V in about 5 seconds, held at this value not less than five seconds, and then gradually decreased to 0. Examinations for evidence of insulation breakdown shall be performed during this test. Reference voltage generators which have once been subjected to this test shall not be subjected to it again.

4.8.5 RF components (VSWR). A standard RF voltage standing wave ratio reflectometer set-up, see Figure 1, with VSWR accuracy of ± 2 percent of indication and frequency accuracy of ± 5 MHz of setting, shall be connected to the input waveguide section of the antenna scanner. The VSWR shall be measured at the frequencies indicated in 3.7.1.5. The measurements shall be made with the antenna scanner in a static position in conical scan (any angular position). The VSWR's shall be determined, for the antenna scanner radiating into space, for conformance to the values specified in 3.7.1.5 for frequencies indicated where the return space reflection is not greated than 0.5 percent.

4.8.5.1 Test procedure for measuring VSWR of input wave guide of radar antenna scanner. Connect the components to the antenna scanner as shown in Figure 1, and measure the VSWR over the band as follows:

(a) Energize the equipment and allow it to warm up.

(b) Disconnect the test set-up at the junction of the coupler and the antenna input.

(c) Connect a standard mismatch (Hewlett Packard (H.P.) X 916 or equivalent) to the detached end of the coupler.

(d) Adjust the ratio meter to read the reflection coefficient stated on the standard mismatch.

(e) Remove standard mismatch and reconnect the antenna scanner.

(f) Measure percent reflection over the frequency band, specified in 3.7.1.5 using the calibrated frequency sweep limits of the oscillator.

* (g) Convert the percent reflections to VSWR using the following relation:

$$VSWR = \frac{1 + \frac{\% \text{ Reflection}}{100}}{1 - \frac{\% \text{ Reflection}}{100}}$$

* 4.9 Mechanical tests. Each antenna scanner shall be lubricated with oil as specified in 3.5.1, and run for not less than twenty-four hours.

The scanner shall be cycled between conical and spiral scan while elevated in steps through -15° to +90°. The average of each operating mode shall be 12 hours. The operation at

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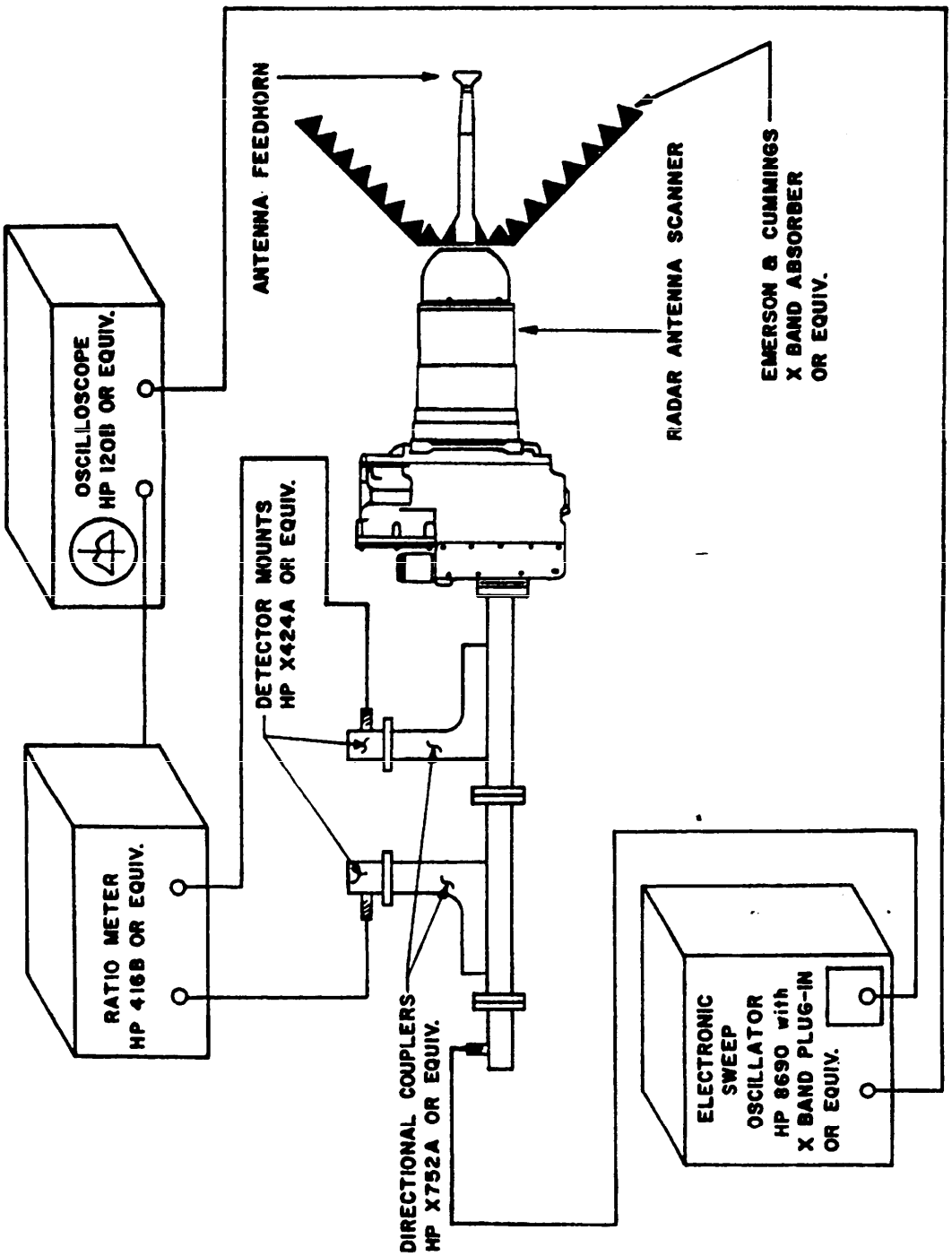


Figure 1. Test Arrangement for Measuring VSWR of the Radar Antenna Scanner

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various elevations shall be in 15° steps returning to 0° between each. The operating period for elevations up to 45° shall be 1 hour and for elevations above 45° shall be 15 minutes. After this run, the antenna shall be checked to determine whether the following requirements have been met:

(a) Nutating speed not less than 1700 RPM.

(b) Direction of nutation for both types of scan as specified in 3.7.2.4.

(c) In conical scan, the circle described by a point in the center of the feedhorn cap specified in 3.7.2.1.

(d) In spiral scan, the distance across the outermost coil described by a point in the center of the feedhorn cap specified in 3.7.2.2.

(e) The concentricity of the circle described in 4.9(c) with respect to the mounting pilot within the tolerance specified in 3.7.2.1.

(f) The oil pump delivering oil to the front bearing assembly as evidenced by the following check: Remove the oil flow inspection plug from bearing retainer, energize antenna scanner, and observe that oil flows from the opening in the bearing retainer.

(g) The antenna scanner showing no signs of oil leakage during the run-in test. Particular attention shall be used when inspecting the front bearing seal for external leaks and the seals of the reference generator shafts, solenoid plunger and rear waveguide tube for internal leaks. No oil seepage at any of these points.

(h) After all of the above testing has been completed and the oil has been drained, the antenna scanner shall be inverted to a vertical position with the feedhorn pointing downward. Visible oil at the bearing seal or on the rotor shaft after ten minutes in this position shall indicate failure to pass this test.

4.10 Environmental tests.

4.10.1 Temperature. The antenna scanner shall be set up for testing in a temperature-controlled room and subjected to the temperature cycles specified in 4.10.1.1 through 4.10.1.3.

4.10.1.1 Storage. The unit shall be subjected to the following temperature cycles:

(a) Reduce room temperature to -40°F and maintain this temperature within ± 5° for at least 2 hours.

(b) Increase room temperature to +160°F and maintain this temperature within ± 5° for at least 2 hours.

(c) Reduce room temperature to +77°F and maintain this temperature within ± 5° for at least 2 hours.

4.10.1.2 Low temperature starting. The antenna scanner shall be de-energized except for the heating and thermostat circuit described in 3.7.1.2. The temperature shall be maintained at 0°F ± 3° for a period of at least 4 hours. The equipment shall then be energized and observed for normal operation within 5 minutes.

4.10.1.3 Operating. The unit shall be observed for continuous operation throughout the following temperature cycle without deviation from the requirements of this specification:

(a) Energize antenna scanner and operate in spiral scan.

(b) Reduce room temperature to -20°F and maintain this temperature within ± 3° for at least 2 hours.

(c) Increase room temperature in steps of 10 degrees F, at 30 minutes per step, until a temperature of 160°F is reached. Maintain this temperature within ± 3° for at least 2 hours.

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(d) Reduce room temperature in steps of 10 degrees F, at 30 minutes per step, until 77°F is reached. Maintain this temperature within ± 3° for at least 2 hours.

4.10.2 Vibration. The antenna scanner shall be subjected to vibration tests in accordance with MIL-STD-167. Type 1 vibration, as specified therein, shall apply.

4.10.3 Degree of enclosure. The waterproof integrity of the antenna scanner shall be determined by the following test: Remove air vent from gearcase top and plug gearcase vent. Fill antenna scanner housing with compressed air to 1-1/2 psi gauge pressure and immerse unit in a six inch head of water for a period of one minute. While maintaining pressure, the housing shall be examined for evidence of escaping air.

4.10.4 Shock. The antenna scanner shall be subjected to shock tests in accordance with classification Type A, MIL-S-901.

* 5. Packaging. Preparation for delivery shall be in accordance with MIL-E-17555.

* 5.1 Preservation. Cover plate assemblies shall be installed on front and rear of the waveguide tube before packing. These prevent foreign material from contaminating the internal radiating surfaces during storage and shipping.

* 5.2 Preparation for shipping. After run-in and testing, drain the lubricating oil. Attach a warning tag to the oil filler plug to refill with oil in accordance with the lubrication instruction plate before operating.

* 5.3 Packing. Each scanner shall be shipped in a reusable metal or wood container. The scanner shall be bolted to a removable wood or metal plate that will permit no movement during transit. Each shipping container will contain two pint cans of lubricating oil and adequately secured to prevent damage during transit. See figures 2 and 3.

* 5.4 Marking. In addition to the markings required by the contract or order, shipping container shall be in accordance with MIL-STD-129.

6. Notes.

* 6.1 Intended use. The AS-515A/SPG Antenna Scanner is intended for use with Radar Sets AN/SPG-34, AN/SPG-50, AN/SPG-53 and AN/SPG-53A and Radar Equipments MK 34 and MK 35. The MX-7232/SPG-53B Antenna Scanner is intended for use with Radar Sets AN/SPG-53B, AN/SPG-53C and AN/SPG-53D.

* 6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) The type of antenna scanner being procured. (AS-515A/SPG or MX-7232/SPG-53B) (See 1.1)

(c) The number of preproduction samples. (See 3.2)

(d) The serial numbers to be used on the identification plates. (See 3.9)

(e) The activity at which preproduction testing will be performed. (See 4.3)

(f) The number of periodic production samples and the method of sampling. (See 4.4)

(g) The activity at which periodic production testing will be performed.

(h) Disposition of failures.

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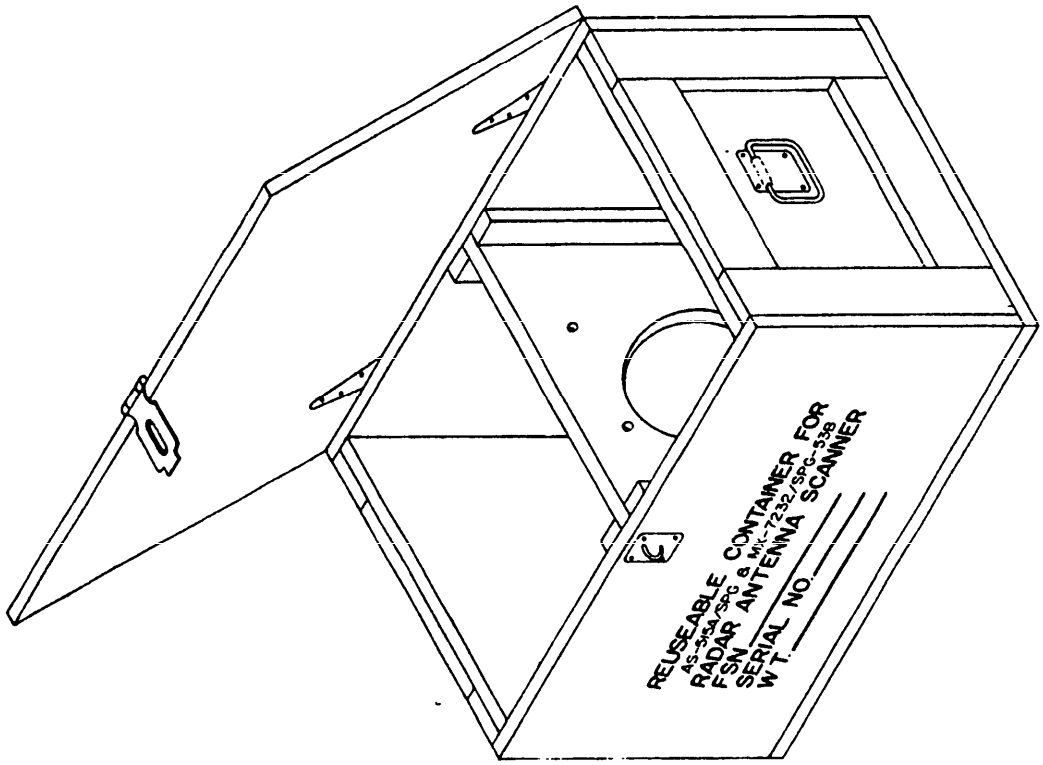


Figure 2. Reuseable Container for AS-515A/SPG & MX-7232/SPG-53B Radar Antenna Scanners

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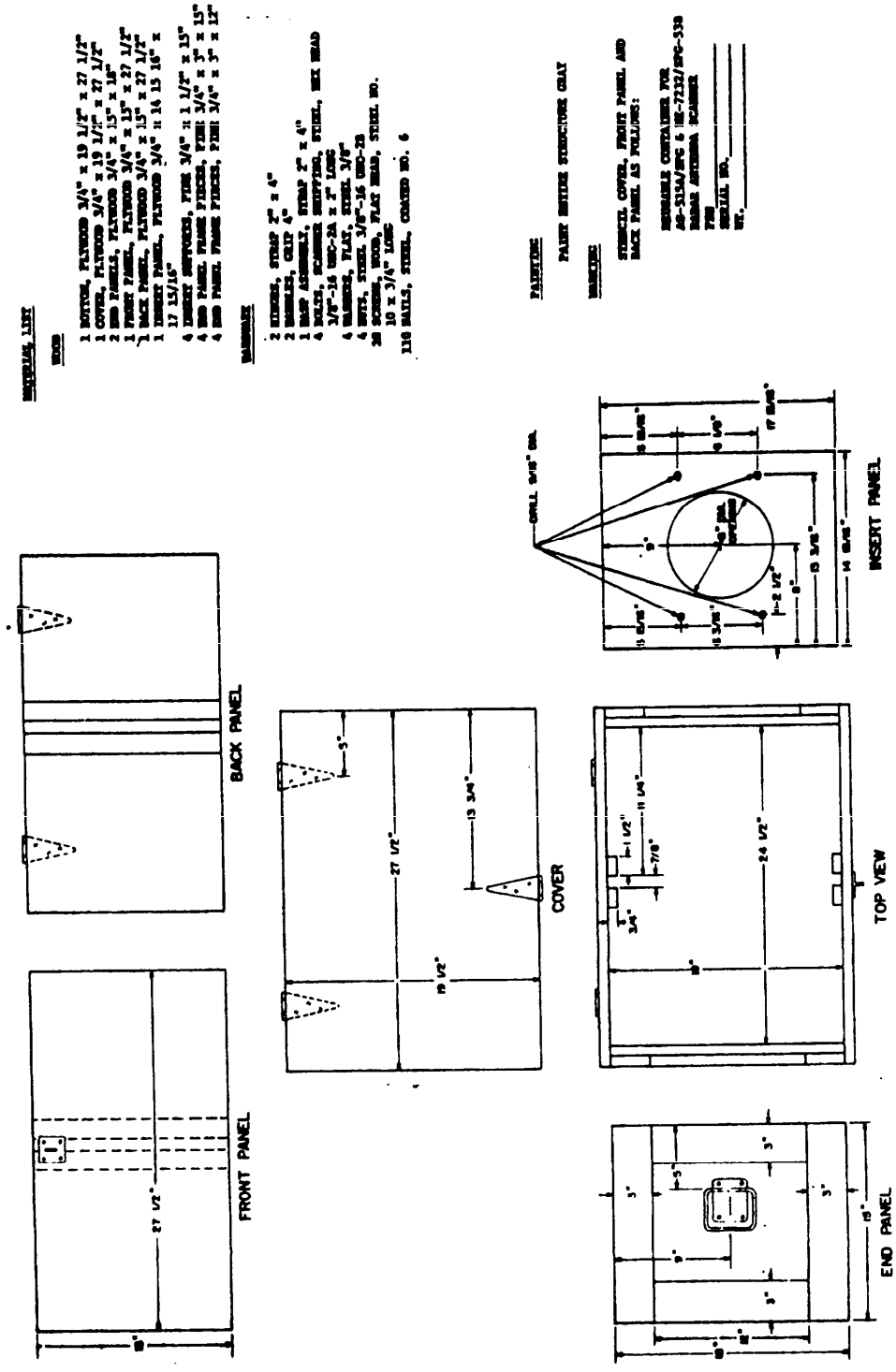


Figure 3. Construction Details, Reuseable Container for AS-515A/SPG & MX-7232/SPG-53B Radar Antenna Scanners

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6.2.2 Contract data requirements. The following paragraphs herein refer to data items which may be required by the contract. Such data will be delivered only when they are specifically itemized on DD Form 1423 incorporated into the contract. The content and format of deliverable data may be governed by specified DID's (Data Item Descriptions).

(a) Non-standard part descriptions. (See 3.5.2)

(b) Design or construction deviations from the drawings. (See 3.6)

(c) Test data. (See 4.3, 4.4, 4.5)

6.3 Cognizance. Technical cognizance of this document is located at the Naval Sea Systems Command Headquarters (SEA-04322D), Arlington, VA.

Custodian:
NAVY-OS

Preparing Activity:
NAVY-OS
Project Number
5985-N461

Note: The margins of this revision have been marked with an asterisk (*) to indicate those paragraphs containing significant changes from the previous revision. Minor changes are too numerous to indicate.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-A-19273B(OS) Antenna AS-515A/SPG and MX-7232/SPG-53B (RADAR ANTENNA SCANNER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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